

Yasemin Ozkut

AI / ML Engineer | LLMs · Agents · Computer Vision · Deep Learning
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Portfolio | LinkedIn | GitHub | Hugging Face

AI/ML engineer who both trains models and builds AI systems on top of them, across LLMs, agents, computer vision, and deep learning, end to end through deployment. Contributing to open-source in personal time.

Education

The Ohio State University Master of Science, Electrical and Computer Engineering Focus: Computer Vision, Machine Learning, Deep Learning	Aug 2024 – May 2026 Columbus, OH, USA
Sabanci University Bachelor of Science, Computer Science and Engineering	Sept 2019 – Feb 2024 Istanbul, Turkey

Skills

Languages: Python (5+ years hands-on), C++, SQL, JavaScript
LLMs & Agentic Systems: LLM/VLM fine-tuning (QLoRA, Unsloth, TRL), LangGraph, multi-agent orchestration, tool-calling, prompt caching, structured outputs (Pydantic), Anthropic API, OpenAI API, ElevenLabs API
Computer Vision: YOLO, OpenCV, object detection, image/video classification, medical imaging (MONAI)
Deep Learning: PyTorch, TensorRT, ONNX, CNN/Transformer, multi-GPU training, quantization, TensorBoard
Frameworks & Tools: Hugging Face, FastAPI, Flask, React, Scikit-learn, Docker, Git, GitHub Actions (CI/CD), CUDA, SLURM, Linux, PostgreSQL, MongoDB, Reachy Mini SDK, Claude Code CLI

Experience

- Computer Vision Engineer Intern**, Ubihere (Early Startup) – Columbus, OH Aug 2025 – May 2026
- Built a configurable React dashboard from scratch, replacing hardcoded per-client builds with one extensible platform supporting client-specific widgets, adopted as the primary client-facing product and used in sales demos.
 - Shipped a research-stage person re-identification modules into live production, building it end to end as a GPU-accelerated Flask microservice (REST API, Docker, NVIDIA Container Toolkit).
 - Cut inference latency 67% by engineering a GPU batching pipeline with parallelized image I/O.
 - Trained YOLOv8 on a multi-GPU SLURM for multi-class vehicle detection (84% mAP50), targeting 60+ city traffic cameras.
- Graduate Research Assistant in ML/CV**, Computer Vision Lab OSU – Columbus, OH Aug 2024 – Aug 2025
- Shipped the open ERDES ocular-ultrasound video benchmark from de-identified clinical data, now 10K+ Hugging Face downloads, with the paper accepted at Nature Scientific Data. [\[GitHub\]](#)
 - Built a two-stage diagnostic pipeline reaching 97.4% accuracy on retinal detachment and 88.2% on macular status, training 59 models across 8 CNN/Transformer architectures on multi-GPU SLURM.
 - Designed a ratio-based top-k temporal pooling layer with L2-norm frame importance scoring that adapts to variable temporal dimensions, eliminating per-architecture manual k tuning.
 - Fine-tuned multimodal LLaMA-3.2-11B-Vision (4-bit QLoRA, Unsloth, TRL) on 100K+ MIMIC-CXR chest X-rays for radiology report generation, cutting loss 79% (BLEU-1 28.1, ROUGE-1 39.5). [\[GitHub\]](#)
- AI/ML Engineer Intern**, DAI-Labor – Berlin, Germany – [\[GitHub\]](#) July 2023 – Sept 2023
- Built an end-to-end real-time grocery detection pipeline, annotating 1,235 custom images to extend a German grocery dataset and training YOLOv8 to 87.2% mAP50.
 - Integrated Keras-OCR to extract German product labels and benchmarked it against YOLOv8 detection.

Projects

- Agentic Business Advisor**, LLM / Agentic Systems, Personal Project – [\[GitHub\]](#) May 2026
- Architected and shipped an async multi-agent LLM API (3 top-level, 6 sub-agents) with LangGraph, FastAPI, MongoDB, for multi-turn, human-in-the-loop business-problem scoping with web-search and structured problem-tree generation.
 - Engineered for cost efficiency with model tiering (Haiku/Sonnet) and prompt caching.
 - Improved output reliability by decomposing logic into single-responsibility agents with task routing, constraining each agent to a focused job, and validating structured outputs against Pydantic schemas.
 - Containerized with Docker Compose and shipped CI/CD (GitHub Actions) with error handling and observability logging.

- Budi Desktop Robot Companion**, Embodied AI, VLM, Team Project – [\[GitHub\]](#) Feb 2026
- **Top 3 Winner (\$1,000), Embodied AI Hackathon 2026** – by Seeed Studio, NVIDIA, Hugging Face, and Pollen Robotics.
 - Built a real-time voice-and-vision productivity assistant on the Reachy Mini robot, running on a Jetson Orin Nano (8GB) with on-device STT (faster-whisper) and VLM (Cosmos Reason2) and Edge TTS for speech.
 - Owned on-device VLM Docker deployment and shipped a voice-triggered, thread-based reminder scheduler and a VLM-based distraction-detection focus mode.
- Judgy Reachy**, Embodied AI, Computer Vision, LLM, Personal Project – [\[GitHub\]](#) | [\[Hugging Face\]](#) Jan 2026
- **5th place of 75, NVIDIA GTC 2026 Golden Ticket Contest** – Hugging Face x Pollen Robotics Reachy Mini Robot track.
 - Built a real-time phone detection pipeline with personality-driven robot feedback using YOLO26m + TensorRT (**2.69x inference speedup**), Llama 3.1-8B (Groq), ElevenLabs TTS, and expressive robot motion.
 - Shipped as an open-source Hugging Face Space with GitHub Actions CI/CD, supporting MuJoCo simulation, physical robot, and standalone WebGPU browser demo (ONNX Runtime + Transformers.js).
 - Forked and used by the Reachy Mini community via an open-source Hugging Face Space.
- Open-Source Contributor**, Hugging Face Reachy Mini (SDK, Conversation App) – [\[GitHub\]](#) Jan 2026 – Present
- Improved LLM tool-calling in the agentic Conversation App with enum-constrained tools, runtime conditional tool exposure, and prompt refinements.
 - Built an AudioManager fusion layer mixing the robot's LLM voice and emotion sound to prevent audio collisions in the Conversation App (open PR).
 - Fixed community app-install failures via metadata persistence and entry-point matching in SDK.
- Agentic Person Re-ID via LangGraph and VLM Reasoning**, Personal Project – [\[GitHub\]](#) July 2025
- Built a LangGraph agent integrating YOLOv8, Qwen2.5-VL-3B, and Qwen2.5-7B for person detection, description, and cross-frame identity assignment.
 - Designed structured JSON prompts for semantic attributes (hair, clothing, accessories) with persistent memory for global ID tracking and explainable, logged matching.
- Serious Game for Children with Cerebral Palsy**, Sabanci University – [\[GitHub\]](#) Feb 2023 – Feb 2024
- Built a Unity/C# balance-physiotherapy game for children with Cerebral Palsy, capturing live Wii Balance Board session data into MongoDB and generating visual progress reports with Python and Flask.

Publications

ERDES: A Benchmark Video Dataset for Retinal Detachment and Macular Status Classification in Ocular Ultrasound
Nature Scientific Data – [\[Website\]](#) | [\[Hugging Face\]](#) | [\[ArXiv\]](#) Accepted for publication, May 2026
Yasemin Ozkut, Pouyan Navard, Srikar Adhikari, Elaine Situ-LaCasse, Josie Acuña, Adrienne A Yarnish, Alper Yilmaz

Activities & Interests

Hiphop Dancer (2015–Present); **Board Member**, SuDance Club, Sabanci University (2019–2023); **Contestant**, Peak Games Unithon (2022); **Tutor**, Civic Involvement Project – 4th-grade students (2020); **Winter Sports Enthusiast** (Skiing, Ice Skating, Snowboarding)